

Certified analytical bounds for hypersonic reachable sets

A measure-theoretic profile decomposition in validated interval arithmetic

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Abstract

The reachable set of a nonlinear control system can be embedded in a space of occupation measures, where the dynamics become a linear constraint and the reachable set is the union of the supports of the admissible terminal measures. This research asks what that embedding yields in a singular limit, for a vector field known only as a set-valued map enclosed in validated interval arithmetic. On the state space nothing is lost—the family is tight—but its weak limit forgets every event that lasts a vanishing time, and a reachable set is a support, not an integral: the limit satisfies the Liouville equation of the limit dynamics only up to a defect. The target theorem identifies that defect. The loss is read on the energy dissipation along the rescaled time axis, where Lions’ concentration-compactness alternative applies as it stands, and it organizes into a profile decomposition: a limit system in a few slow coordinates, one boundary-layer bubble per concentrating event, entering as a jump, and a remainder. The statement aimed at holds at fixed ε : the reachable set lies in that of the limit hybrid system, thickened by a ball whose radius is an explicit function of ε . Compactness gives the structure and no constants. Those are to come from direct estimates—the energy identity, a matching estimate at each pass, and averaging over the oscillations that do not concentrate, where frequencies that cross are the critical difficulty—and are required in closed form, in the physical constraint set and the aerodynamic interval widths. Hypersonic entry supplies the instance: a six-degree-of-freedom vehicle whose thin-atmosphere limit furnishes the singular perturbation, whose exo-atmospheric arcs and equilibrium glide furnish the limit system, and whose atmospheric passes furnish the bubbles. The aim is a bound whose dependence on the data is explicit: valid uniformly over the entire admissible family, in the minimal physical state space, and re-evaluated rather than re-solved when the vehicle or the corridor changes. The central theorem is open, and this document claims no result: it states the problem, the statement aimed at, and the order in which it is to be attacked.

1 The analytical problem

The object to be bounded is the reachable set of an atmospheric entry vehicle—every state it can be driven to under admissible control—which is what a safety analysis needs enclosed. The flight-mechanics specifics are deferred: [Section 3](#) states why entry is the right instance and what its state space looks like. What matters first is that the problem is an analytical one before it is an aerospace one, and that the existing methods fall short of it on two independent counts: the dimension of the state, taken up with the application in [Section 3](#), and the provenance of the guarantee, in [Section 1.2](#).

1.1 Reachable sets as occupation measures

Embed the flight dynamics in a space of measures. Along a trajectory of $\dot{x} = f(x, c, t)$, with the state x in the corridor $\mathcal{X}$ and the control c in $\mathcal{U}$, the time spent in each set defines an occupation measure μ on $[0, T] \times \mathcal{X} \times \mathcal{U}$, and the nonlinear flight mechanics become *linear*: the Liouville equation is a linear constraint tying μ to an initial measure and to a terminal measure μ_T , and the reachable set at time T is the union of $\text{supp } \mu_T$ over the measures that satisfy it. No single measure bounds it. The outer bound comes from the dual: a pair of functions (Φ, Ψ) , the first non-decreasing along every admissible trajectory and at least one on the initial set, the second majorizing it at time T , encloses the reachable set in the superlevel set $\{\Psi \geq 1\}$. This is the occupation-measure program of Lasserre, Henrion and collaborators [[1](#), [2](#)], developed for regions of attraction and invariant sets [[3](#), [4](#)] and, for hybrid systems, for reachable sets and optimal control, where a measure on each guard carries the mass from one mode into the next [[5](#), [6](#)]. The embedding buys linearity at the price of a relaxation, and the price is known: a measure that satisfies the Liouville constraint is generated by a family of trajectories of the *convexified* inclusion $\dot{x} \in \text{conv } f(x, \mathcal{U}, t)$, not by a single trajectory of the original system [[3](#), Lemma 3]. The relaxed reachable set contains the true one, so an outer bound on the first is an outer bound on the second, which is the direction needed here.

The same relaxation admits the interval envelope. Because the true aerodynamics are transcendental and known only to within an envelope, they are enclosed in intervals, and a vector field selected from an interval is a set-valued map; the resulting differential inclusion is the object the measure must describe, not the nominal field [[7](#), [8](#)]. Written with the selection as an additional box-valued input, the inclusion is a control system with a bounded disturbance, and at fixed ε its Liouville constraint is the standard one. What is not standard is carrying the outer approximation through the singular limit of [Section 2](#), uniformly over the selections. Separating time scales is itself a standard device of flight mechanics [[9](#)], and the classical reduction, which sets $\varepsilon = 0$, requires the fast motion to settle. Occupation-measure theory for inclusions that does not require it already exists, and it is the nearest prior art: it defines an occupational invariant measure without assuming solution uniqueness, which is the same move made here, and lets the slow dynamics be driven by the invariant measures of the fast [[10](#)]. What it assumes is worth stating plainly, because it locates where the entry problem leaves it. The linear growth condition on the field serves only to exclude finite-time blow-up, and is automatic on a bounded corridor. The singular-perturbation theorem, however, assumes a uniform bound on the fast state along the whole ε -family [[10](#), [11](#)], and the natural fast state of entry is the altitude measured in scale heights: on a corridor of fixed altitude extent it ranges over an interval of length proportional to $1/\varepsilon$, the aerodynamic terms grow exponentially in it, and a skip that rises through a fixed altitude carries it out of every bounded set. The hypothesis holds for motion that stays within a bounded number of scale heights of the glide, and it fails exactly where a pass concentrates; [Section 2.1](#) says which is which.

Objective 1: Interval embeddings uniform through the singular limit

Question. What do the occupation measures of the interval inclusion $\dot{x} \in \mathcal{F}_\varepsilon(x, c, t)$ converge to as $\varepsilon \rightarrow 0$, and what linear constraint does the limit satisfy?

To establish. At fixed ε the inclusion is a control system with a box-valued disturbance and its Liouville constraint is standard. So are the bounds that hold uniformly in ε : on mass, on support and, through the energy balance, on dissipation. Two things are not. The first is a certified upper and lower envelope of the atmospheric density, which fixes $\mathcal{F}_\varepsilon$ concretely. The second is the limit of the constraint itself. A weak-* limit of the measures satisfies the Liouville equation of the limit flows only up to a defect, carried where the dissipation concentrates. The claim to be proved is that this defect is a jump term, on a measure supported on the graph of the pass map, so that the limit satisfies the Liouville equation of the hybrid system of [Section 2.4](#), in the form with guard measures [\[5, 6\]](#), uniformly over the selections. Where nothing concentrates the limit is of the kind the existing theory describes, with the oscillation about the glide represented by an invariant measure [\[10\]](#).

Success. The limit constraint stated for the full inclusion, with each hypothesis either verified on the entry corridor or named as an assumption; and its dual, a pair (Φ, Ψ) non-decreasing along the flows and across the jumps of the limit system, as the form in which the outer bound of [Section 1.1](#) survives the limit.

1.2 The provenance of the guarantee

A polynomial barrier whose coefficients come from an interior-point solver has analytical *form* but inherits the solver’s accuracy [\[12\]](#), and the documented failure modes of numerical semidefinite solvers include the loss of strict feasibility on which the repair schemes depend [\[13\]](#). A solver’s output can be made into a certificate: rounded to rational coefficients and re-verified exactly, it is a proof [\[14\]](#). The obstacle is scale, not principle. The rounding needs a strictly feasible Gram matrix, which a tight bound does not have, and exact rational certificates carry a documented bit-complexity whose exponent grows with the number of variables [\[15, 16\]](#)—a cost the thirteen-state field of [Section 3](#) does not invite as a main line.

The program therefore fixes an order of evidence. The certificate itself is a closed-form bound: the constants of [Section 2](#), derived from the decomposition and evaluated over the corridor in validated interval arithmetic. Behind it are to stand symbolic checks of the derivations those constants rest on, made against the same definition of the dynamics that is evaluated numerically. A moment-sum-of-squares computation on a reduced model is the third tier: an independent cross-check, to which the exact-certification machinery above belongs, and never the source of the guarantee.

Interval arithmetic is the setting in which a guarantee has meant, for sixty years, an enclosure rather than an approximation. It is rigorous in floating point: every operation is rounded outward, so the computed interval contains the exact result, and nothing is gained for soundness by exact rational arithmetic, which cannot in any case evaluate an exponential or a sine [\[17, Ch. 3\]](#). Two distinct losses have to be managed, and they are often run together. The first is the *dependency problem*. Interval arithmetic is inclusion isotonic—the property that makes an enclosure valid at all—but not every law of real arithmetic survives the move to intervals: distributivity weakens to subdistributivity, and repeated occurrences of a variable are treated as independent [\[17, Chs. 4–5\]](#). The second is the *wrapping effect*: the image of a box under a flow is not a box, and re-enclosing it at every step admits points that no trajectory reaches [\[17, Ch. 10\]](#). Refining the step does not help. Under interval Euler steps of size h , re-boxed at each, a rigid rotation inflates a box by $(1 + h)^{2\pi/h}$

per revolution, which *increases* to $e^{2\pi} \approx 535$ as $h \rightarrow 0$; what controls wrapping is the representation of the set—Lohner’s QR factorization, zonotopes, Taylor models [18, 19]. Applied interval analysis supplies the machinery the evaluation rests on, and it is deliberately conservative: certified set inversion, contractors, and validated integration [20].

Question. Can the closed-form constants, the pass enclosures they do not cover, and the reachable set of the limit system of Section 2.4 be evaluated over the whole corridor with every rounding error accounted for?

To establish. The evaluation, as a computer-assisted step of the proof: the transcendental field and its Jacobian enclosed with outward rounding, the corridor constraints contracted, and the passes integrated with validated error bounds. The limit system is itself a reachability problem, in the few slow coordinates and over every bank history: it is to be enclosed by comparison bounds in closed form where they exist, and by validated set propagation where they do not.

Success. Every inequality the certificate asserts is discharged by a computation that returns either a proof or an explicit failure to decide at the working precision, and the dynamics it evaluates are the dynamics that are checked symbolically.

2 The singular limit and its profile decomposition

This section carries the new content. The instance supplies a small parameter—in the entry problem the thin-atmosphere ratio $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$, where H_s is the atmospheric scale height and R_e the planetary radius, written ε where the other small parameters are not in play—and the question is what the occupation-measure family does in that limit.

2.1 Loss of compactness along the time axis and the defect measure

On the corridor nothing is lost in the usual sense. The set $[0, T] \times \mathcal{X} \times \mathcal{U}$ is compact, so the occupation measures—which live in the cone $\mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U})$ of nonnegative Radon measures, topologized weakly-* as the dual of $C([0, T] \times \mathcal{X} \times \mathcal{U})$ —form a tight family and converge along subsequences. The limit is nevertheless the wrong object, because it forgets the atmospheric passes (Fig. 2.1). How much it forgets depends on which limit is taken, and that has to be said first.

A pass entered at a flight-path angle γ_e turns the velocity through about $2\gamma_e$ and sheds a share of the kinetic energy of order $\gamma_e/(L/D)$, since lift cannot be had without drag. It crosses a scale height in a time $H_s/(V\gamma_e)$, of order $\varepsilon_{\text{atm}}/\gamma_e$ against the slow time, and it levels out where the lift is $V^2\gamma_e^2/(2H_s)$, of order $\gamma_e^2/\varepsilon_{\text{atm}}$ against gravity. The entry angle therefore enters through $\gamma_e/\sqrt{\varepsilon_{\text{atm}}}$ —as in the classical Chapman–Loh variables, it is $\sqrt{\varepsilon_{\text{atm}}}$ and not ε_{atm} that sets the scale [22, 23]—and two limits have to be told apart. With $\gamma_e/\sqrt{\varepsilon_{\text{atm}}}$ held fixed, a pass lasts a time of order $\sqrt{\varepsilon_{\text{atm}}}$, sheds an energy of the same order, and loads the vehicle by a fixed multiple of what the glide does. This is the limit of a corridor whose dynamic-pressure and load limits are held fixed, and in it nothing concentrates: the passes are many and slight, the troughs of an oscillation about the glide. With γ_e held fixed, a pass lasts a time of order ε_{atm} and sheds a fixed share of the energy, under loads that grow like $1/\varepsilon_{\text{atm}}$. It carries occupation mass of that order, and none in the limit, while its share of what distinguishes one trajectory from another—the energy dissipated, the turn of the velocity vector—does not vanish with it. The physical case lies between the two. $\sqrt{\varepsilon_{\text{atm}}} \approx 0.034$ is about two degrees and entry angles are a few degrees, so a pass need be neither short against the arc that follows it nor distinct in scale from the oscillation about the glide. The statement aimed at

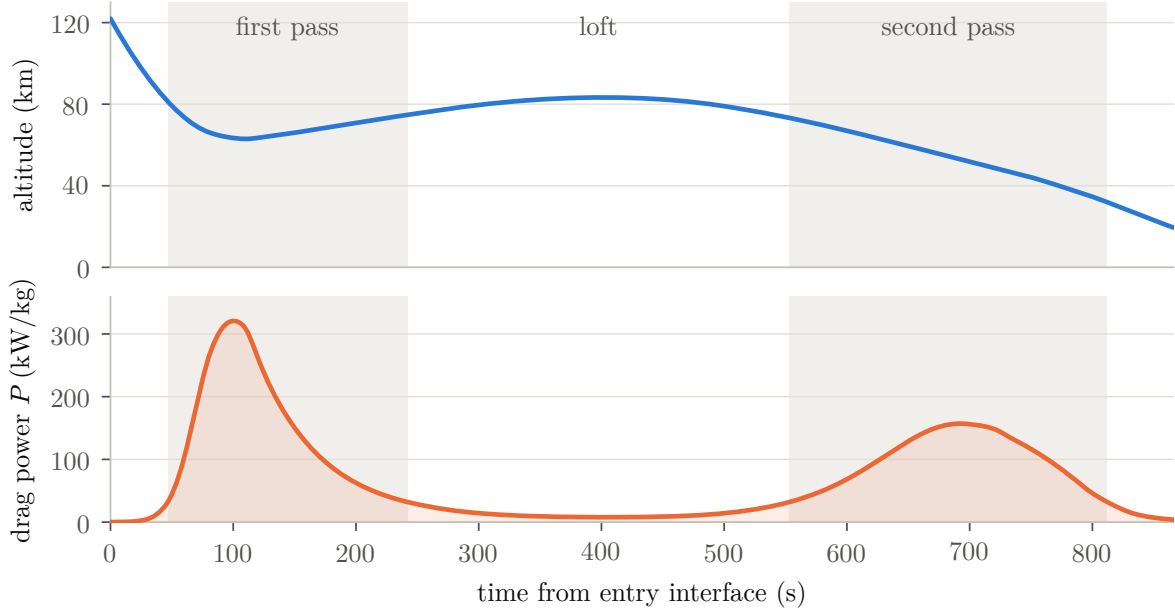


Figure 2.1: A skip entry at the physical scale height, by way of example: the lunar return of Artemis I, simulated from the published entry state [21]. Above, altitude. Below, the drag power per unit mass P , the rate at which mechanical energy is dissipated. The shaded intervals are the two atmospheric passes, on which P exceeds a tenth of its peak; between them the vehicle lofts and dissipates little. An occupation measure weighs every interval by its length. What distinguishes one trajectory from another is decided in the shaded ones, and in the limit they have no length.

is therefore one at fixed ε , with a threshold that decides which passes are resolved one by one, and the two limits are the ends it has to cover.

A reachable set is a support, not an integral, and what has no mass still counts. That alone does not set this problem apart from the limit theory of singularly perturbed control, whose value functions already carry a terminal cost on the slow state [11]. What sets it apart is the hypothesis under which that theory works: the fast state, and the trajectories that approximate an optimum, stay in a bounded set, and a pass that concentrates is what that excludes. The loss has to be read where it occurs, along the fast-time axis $\tau = t/\sqrt{\varepsilon_{\text{atm}}}$, on which the horizon $[0, T/\sqrt{\varepsilon_{\text{atm}}}]$ exhausts the half-line and translations act non-compactly.

Between passes the vehicle is outside the sensible atmosphere, on an arc governed by gravity alone; once captured, it stays near the equilibrium-glide balance $\Lambda = 1$. In the 3-DOF (point-mass) approximation this is the scalar condition $L \cos \sigma / [m(g - V^2/r)] = 1$; in the full 6-DOF formulation the lift component $L \cos \sigma$ is not an independent input but is determined by the attitude quaternion through the body-to-LVLH rotation, so the balance is a single scalar constraint on the 13-dimensional translational-rotational state, and the slow set is its intersection with the trim and quasi-static-deflection conditions of the two faster motions—one constraint set for each. The balance is taken with planetary rotation and oblate gravity retained, and the two do not enter on equal terms: flying east at entry speed the Coriolis term is tens of percent of the net acceleration $g - V^2/r$ that the lift has to balance, while the J_2 terms are a percent of it at most. The scalar $\Lambda = 1$ is the reduction under spherical gravity, no rotation and small flight-path angle.

The loss is read on a *defect measure*, and the object has to be chosen so that the passes carry mass. The part of the occupation-measure limit that lies off the glide cannot serve, since the

passes have none. The energy dissipation can. Along every trajectory of the inclusion the specific mechanical energy in the rotating frame, $\mathcal{E} = \frac{1}{2}V^2 + \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$, changes at the rate $\dot{\mathcal{E}} = -P$, where $P = DV/m \geq 0$ is the drag power: neither the lift nor the Coriolis force does work. The total dissipation $\int_0^T P dt \leq \mathcal{E}_{\max} - \mathcal{E}_{\min}$ is therefore bounded uniformly over controls, over selections of the interval field, and over ε . Let

$$\nu_\varepsilon(\tau) = \frac{\sqrt{\varepsilon} P(\sqrt{\varepsilon} \tau)}{\int_0^T P dt} \quad (2.1)$$

be the dissipation density on the fast-time axis, normalized to unit mass and extended by zero. It is proposed as the defect measure. Where it concentrates is where the limit of the occupation measures can stop satisfying the Liouville equation of the limit flow, and the mass it puts there should bound the size of the failure, because the aerodynamic impulse over any interval is bounded by the energy dissipated in it. Along any sequence $\varepsilon_n \rightarrow 0$, with a trajectory chosen for each, the ν_ε are nonnegative densities of fixed mass on the line, which is exactly the hypothesis of Lions' concentration-compactness lemma [24, Lemma I.1], and here all three of its alternatives mean something:

- *vanishing*—no window of fixed fast-time width holds a fixed share of the dissipation: the glide and the oscillation about it, on which the drag is bounded and such a window holds a share of order $\sqrt{\varepsilon}$;
- *compactness*, up to translation—one pass carries the dissipation;
- *dichotomy*—the dissipation splits between pieces whose fast-time separation diverges: two passes a positive time apart, or a pass followed by a glide.

Which of the three occurs is decided by the scaling above. With the loads held fixed the drag power is bounded and every sequence vanishes. Concentration needs an entry angle that stays large against $\sqrt{\varepsilon}$. Then translations along the fast-time axis separate the passes, and it is the dilation half of the principle—the limit-case modification [25]—that resolves each one at its own width, since a pass shorter than $\sqrt{\varepsilon}$ is a Dirac mass at that scale. The trichotomy is Lions' lemma applied as stated. That the vanishing part is the glide and the concentrated pieces are the passes is asymptotics still to be argued, from the inner and outer expansions of the classical theory [26].

Measures that keep track of concentration are not new to this subject. An unbounded control that enters the dynamics affinely is relaxed to a measure in time whose atoms are jumps of the state, and oscillation and concentration of a control sequence have been carried together by the measures of DiPerna and Majda [27, and the references there]. There the concentration is put in by the control. Here the control is bounded and the limit produces it, through a force that depends on the state, so what a jump does is the solution of an inner problem; bounding the impulse alone gives a limit that is sound and far too large.

2.2 Skeleton, bubbles, residual

Rather than restore compactness, the program extracts the profile decomposition that iterating the dichotomy hands it:

- a *skeleton*, low-dimensional—what survives the limit: the exo-atmospheric arcs between passes, which are conics up to the oblateness correction, followed by the equilibrium glide, a graph over (r, V) in the corridor subspace once the balance is solved for the vertical lift fraction. For the glide a rigorous characterization with quantitative accuracy estimates already exists [28], and it treats the glide as a *regular* perturbation: the exact solution is not attracted to the zeroth-order one, a point that returns in Section 2.4;

- one *boundary-layer bubble* per atmospheric pass, the analogue of a concentrating profile: the pass rescaled to its own width. Its net effect on the slow coordinates is a jump. The exit states that the bank history and the selections can produce from one entry state form a compact set, and the map that assigns it, the *pass map* J , is the object to be enclosed. Orthogonality is read off the scale-and-core separation condition that governs concentrating profiles generally [29]. The decomposition of skip entry into an aerodynamic inner region and a gravitational outer one is classical [22, 26], and it has been carried to second order and to three dimensions [30]. Those are expansions of one trajectory, with the error assessed by example. What is not classical is a statement uniform over a family of trajectories, with an explicit remainder at fixed ε ;
- a *residual*: what the finitely many profiles do not capture. In a profile decomposition the remainder is not compact and is not small in the conserved norm; it is small in a weaker one, and only as the number of extracted profiles grows [31, 32]. Here the weaker norm is the largest share of dissipation held by any fast-time window: on a glide the residual carries most of the dissipation, and almost none of it in any one window. The quantitative bound the program needs is a different statement, a distance in state space, and it is taken up in Section 2.4.

The two kinds of statement are kept apart, because they come from different arguments. A decomposition obtained by compactness is a statement about subsequences, and what it gives uniformly it gives by contradiction: it says which pieces can occur, and that nothing else can, and it produces no constant. The wave-equation theory is explicit about this. The uniform bound that follows from its decomposition comes with a constant its authors say they cannot describe [31, Corollary 2]. Here compactness is asked only for the structure. The constants come from direct estimates: the energy identity, the bound it gives on the aerodynamic impulse, a matching estimate at each pass, and averaging over the oscillations that do not concentrate. Each of those is an inequality with a constant that can be written down, and writing them down is [objective 6](#).

The number of bubbles is bounded *a priori*, and the bound comes from the same monotone quantity. A pass that dissipates at least δ of specific energy can occur at most

$$K_\delta \leq \frac{\mathcal{E}_{\max} - \mathcal{E}_{\min}}{\delta} \quad (2.2)$$

times, for every control and every selection of the interval field. Passes below the threshold are left to the oscillation about the glide and to the residual, which is the device of the profile-decomposition literature: only finitely many profiles exceed any given size. The integrated heat-load budget gives the count its physical reading—a trajectory can only afford so many skips [26, 33]—but it routes through a stagnation-point heating correlation, and the energy budget does not.

Enclosing a pass is where the wrapping effect has to be met. For linear systems it can be avoided altogether: support-function methods propagate each direction independently, so successive over-approximations do not compound, and what remains is a discretization error of first order in the time step [34]. Nothing comparable is available for a nonlinear pass.

Objective 3: Enclosures of the pass maps

Question. Can the net effect of one atmospheric pass on the slow coordinates be enclosed tightly enough that a sequence of K_δ passes does not exhaust the corridor?

To prove and compute. Wrapping in a pass has a specific source. Trajectories of a set enter the layer at different times, the set is sheared along the flow, and re-enclosing the sheared set in a box at every step admits points that no trajectory reaches. The remedy that fits the decomposition is to enclose the *pass map*—entry state to exit state, in the rescaled time of the profile—as a single object, a zonotope or Taylor-model enclosure of a map rather than a tube of boxes, and to compose pass maps with the outer flow.

Success. A bound on the growth of the enclosure per pass, explicit in the interval widths, and its verification on a multi-pass corridor against sampled trajectories.

2.3 Orthogonal scales, and what does not concentrate

Two things have to be proved before the decomposition is a bound, and they differ in kind.

The first concerns the passes. In the wave-equation theory two profiles are orthogonal when their scales diverge or, at equal scale, when their centers separate by a diverging number of widths [31, (1.18)]. Here the scale of a profile is the width of a pass and its center the time at which the pass occurs, so passes at distinct times are orthogonal with nothing to prove. The dissipation is a measure, so it splits *additively* across orthogonal profiles; that is the analogue of the energy identity of that theory, and it costs nothing. The state does not split additively, because an ordinary differential equation composes: the state leaving one pass is the datum of the next arc. What has to be proved is the analogue of the nonlinear statement of that theory, that the contributions of separate profiles do not interact up to a small error [31, (1.26)]. Here it reads: the map from entry data to exit data factors through the individual pass maps and the flows between them, up to a remainder that is explicit and uniform over the family. This is a matching estimate between inner and outer solutions. The classical theory supplies its form, for one trajectory [26, 30], and its small parameter is the one of Section 2.1: inside a pass gravity is negligible against the aerodynamic force in proportion as $\gamma_e^2/\varepsilon_{\text{atm}}$ is large, which is why a pass below the threshold cannot be treated as a jump at all. For the passes the argument worth transporting from the wave-equation theory is the constructive one: splitting a bounded sequence into an oscillatory and a singular part at each scale forces those scales to be orthogonal, and iterating extracts a pairwise orthogonal family, so that orthogonality is produced by the construction and not assumed by it [31].

Objective 4: Composition of passes

Question. Does a trajectory with several passes factor through the pass maps and the flows between them, with a remainder that is explicit and uniform over the family?

To prove. A non-interaction theorem for passes separated in time, for every selection of the interval field, with a matching error at each pass whose constant is in closed form. The statement has a range of validity, $\gamma_e^2/\varepsilon_{\text{atm}}$ above a threshold, and through the relation between the turn of a pass and the energy it sheds that threshold is the δ of Eq. (2.2). Passes below it are not composed but averaged.

Success. The factorization with its remainder, for the translational state under interval disturbances, verified against sampled multi-pass trajectories.

The second concerns what does not concentrate. The attitude and the elastic motions are not events. They are oscillations, at the short-period attitude and first structural periods T_{att} and

T_{ela} , measured against the slow time $T_{\text{slow}} = \sqrt{R_e/g}$ by $\varepsilon_{\text{att}} = T_{\text{att}}/T_{\text{slow}}$ and $\varepsilon_{\text{ela}} = T_{\text{ela}}/T_{\text{slow}}$, and they persist for as long as there is air. So does the oscillation about the glide, whenever the loads are bounded. In a profile decomposition motions of this kind are not profiles. They stay in the remainder, which the wave-equation theory is careful to say is not small in energy, since it contains every wave that concentrating profiles cannot represent [31, p. 136]. The tool for them is averaging [35, 36]: each oscillation is carried by a slowly varying amplitude, and the slower dynamics see it through its mean. Two things stand between that tool and a bound. An averaging theorem gives the error to an order, and an order is not a bound; the standard text says as much of its own estimates [35, Sec. 10.1]. And with several oscillations the theorem holds only away from resonance, while the frequencies here are not constants. That of the attitude grows with the square root of the dynamic pressure, so it rises from nothing at the start of every pass, and all of them are known only to within the aerodynamic intervals. Where two frequencies cross, averaging over both phases fails in a zone about the resonance; the treatment there is itself an inner expansion matched to an outer one, and a complete theory is not available even for a single smooth field [36, Chs. 7–8]. The results that pass through a resonance without resolving it hold, where capture is possible, outside a set of initial conditions of small measure, and an outer bound can discard nothing. This is the obligation for which no theory is at hand, and it sits on the critical path.

Question. Can the oscillations that do not concentrate—about the glide, of the attitude, of the structure—be averaged with an error that is explicit, uniform over the selections of the interval field, and valid for every trajectory, including those that pass through a resonance?

To prove. An averaging theorem with constants. The amplitude of each oscillation is carried as a slow coordinate of the limit system of Section 2.4 or enclosed in the interval field, and the theorem bounds the distance between the slow coordinates of the full system and those of the averaged inclusion, in closed form. The interval widths enter twice: as a disturbance of each oscillation, and through the frequencies, which depend on the uncertain coefficients, so that the ordering of the frequency bands is itself a hypothesis to be verified over the corridor. Where two bands cross, the statement is for the coupled pair, carried jointly through the crossing, with the width of the resonance zone and the change of amplitude across it enclosed.

Success. The theorem for the oscillation about the glide and for the attitude, with the remainder in closed form; the elastic modes either included or enclosed as a widened interval, as in the fallback of Section 4.

2.4 The certified bound

Skeleton and bubbles together are not a surface with corrections but a system of its own, with flows and jumps, and the target is stated against it. In the slow coordinates the *limit system* $\mathcal{H}_\delta$ has two flows and one jump. On an arc the state moves under gravity alone. On the glide it moves along the balance $\Lambda = 1$, under the admissible bank histories and the selections of the interval field, and it carries one coordinate more: the amplitude of the oscillation about the balance. A pass that dissipates more than the threshold δ of Eq. (2.2) is a jump. The state that enters it is sent to the set $J(x)$ of states that can leave it, onto an arc or onto the glide, with the time and the range the pass consumes included, and a trajectory makes at most K_δ of them. The system has the form the limit dictates, and its data—the pass map, the glide—are those of the given ε . The target statement is

$$\mathcal{R}_\varepsilon(T) \subseteq \Pi(\mathcal{R}_{\mathcal{H}_\delta}(T)) \oplus \mathbb{B}(\varrho(\varepsilon, \delta)), \quad (2.3)$$

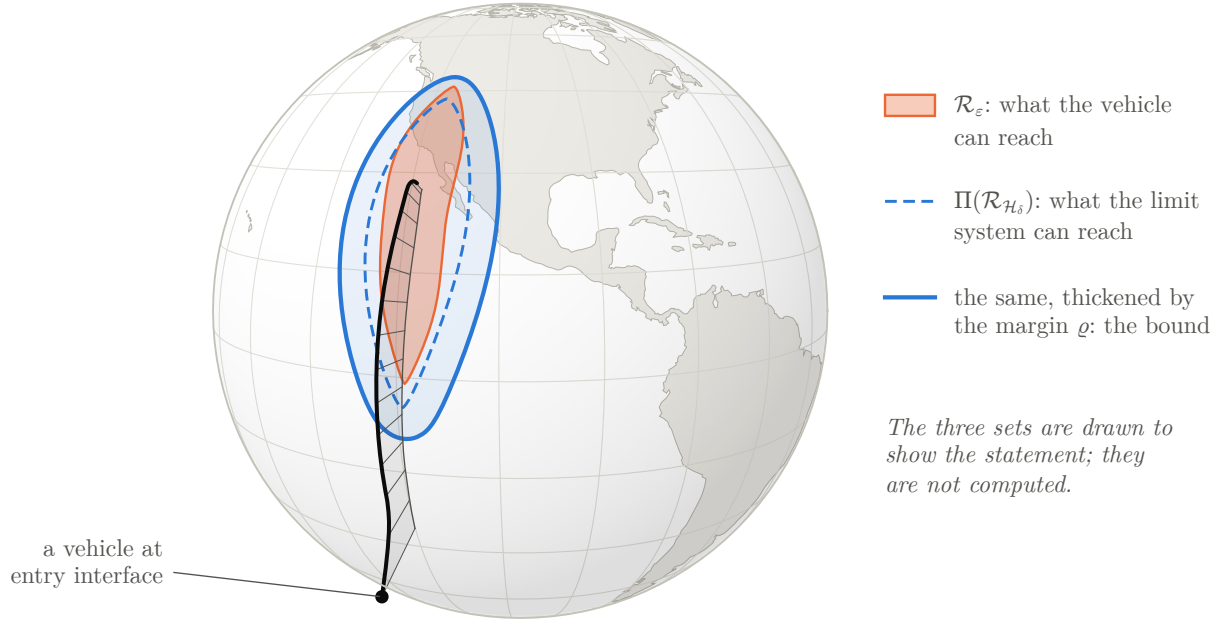


Figure 2.2: The target statement Eq. (2.3), drawn on the ground. The trajectory is the example of Fig. 2.1, with altitude drawn twelve times too large and a post each minute. The three sets are an illustration and are not computed: what the vehicle can reach, what the limit system can reach, and the second thickened by the margin, which is the bound. The first may leave the second, but by no more than the margin.

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory. Here $\mathcal{R}_\varepsilon(T)$ is the set of slow states reachable at time T from the initial set, under every admissible control and every selection of the interval field, and $\mathcal{R}_{\mathcal{H}_\delta}(T)$ is the corresponding set for the limit system. Π reads a state of the limit system back in the coordinates of the full one: it is the identity on an arc, and on the glide it returns the altitudes and flight-path angles that an oscillation of the carried amplitude visits. The sum $\oplus$ is the Minkowski sum, and $\mathbb{B}(\varrho)$ is the ball of radius ϱ in the corridor metric, a weighted supremum norm in which altitude is measured in scale heights and each remaining slow coordinate in units of its corridor width. Passes compose in sequence, as jumps do, and their effects are not added: an arc that meets the atmosphere has no continuation without the pass. Figure 2.2 draws the statement on the ground.

Equation (2.3) is a statement about sets. As a statement about measures it is the claim of objective 1, that the occupation measures converge to measures satisfying the Liouville equation of $\mathcal{H}_\delta$, with the mass that crosses each pass carried by a measure on the graph of J . And the dual pair (Φ, Ψ) of Section 1.1, non-decreasing along the two flows and across the jump, is the form in which an outer bound on $\mathcal{R}_{\mathcal{H}_\delta}(T)$ is certified.

That the glide carries an amplitude is forced by the dynamics. Write z for the altitude above the glide in scale heights and $G = g - V^2/r$ for the acceleration the lift has to balance. Lift falls off exponentially with altitude and G does not, so to leading order in ε , at frozen speed and with the damping left out,

$$\ddot{z} = \frac{G}{H_s} (e^{-z} - 1). \quad (2.4)$$

The glide is its equilibrium. The phugoid, the oscillation of altitude against flight-path angle about the glide, is its small oscillation, at the frequency $\sqrt{G/H_s}$, which grows like $\varepsilon^{-1/2}$. A skip is its

large one: a rebound from the exponential wall below and a ballistic arc under $-G$ above, which is how the classical theory describes skip paths, as oscillatory paths of large amplitude [26]. An oscillation that crosses the glide altitude at a flight-path angle γ_0 has an amplitude fixed by γ_0^2/ε , the parameter of the scaling in Section 2.1. The two limits told apart there are the two ends of one family, and the threshold δ is a cut in the amplitude: an oscillation above it is resolved pass by pass, and one below it is averaged.

What the equation leaves out decides the analysis. With L/D the lift-to-drag ratio and σ the bank angle, the damping rate of the phugoid is $g/[V(L/D)\cos\sigma]$. It does not depend on ε , while the frequency grows like $\varepsilon^{-1/2}$. In the limit the motion normal to the glide is an undamped oscillation on its own timescale: the glide balance is not a normally hyperbolic slow manifold in the sense the geometric theory requires [37], and no boundary-layer estimate of Tikhonov type brings a trajectory onto it. Nor is the amplitude small. A change of bank moves the glide altitude by a fixed fraction of a scale height however thin the layer, and a bank history in step with the oscillation pumps it, as one opposed to it damps it. Guidance laws use the second [38], and a reachable set has to allow for the first. The amplitude is therefore a coordinate on which the control acts, and its range belongs to $\mathcal{R}_{\mathcal{H}_\delta}$, not to the residual.

The residual $\varrho(\varepsilon, \delta)$ collects what the limit system does not carry, and it has three parts: the lag of a trajectory behind the descending glide, the matching error at each of at most K_δ passes, and the averaging error of the oscillations. It and the count K_δ are to be derived as explicit, closed-form expressions in the corridor bounds and the aerodynamic interval widths, independent of any numerical optimization. One piece is immediate: the count is Eq. (2.2). The rest is to be derived, and what to expect of the first part can be said. The lag, in scale heights, should be of order $\sqrt{\varepsilon}$, the order of the damping ratio of the phugoid, with a constant in the speed, the lift-to-drag ratio and the bank that degenerates as V approaches circular speed, which is where the glide gives way to arcs and passes. It would be the ringing amplitude of a trajectory started level on the glide, and no more than that: it says nothing of a trajectory that leaves a pass, which rings with whatever amplitude the pass left it, for a time comparable to the entry. That amplitude is what the limit system carries.

In the spherical-LVLH coordinates the altitude and speed bounds of the corridor are box constraints on (r, V) . The heating, dynamic-pressure and load limits are not: each is a curve $h \geq h_{\min}(V)$. Each is, however, monotone in altitude and in speed at fixed attitude, so contracting a box against it returns the exact interval hull of the feasible part, and the interval widths that enter ϱ and K_δ inherit that tightness. Requiring the constants in closed form is not what makes the bound a proof; a verified solver output is one too (Section 1.2). It is what makes the dependence on the corridor and on the interval widths explicit, so that the bound is evaluated again, and not solved for again, when either changes.

Objective 6: Closed-form constants

Question. What are K_δ and $\varrho(\varepsilon, \delta)$ as functions of the corridor bounds, the initial set and the aerodynamic interval widths?

To derive. The count from the energy budget, with the threshold δ chosen against the residual it leaves behind: lowered, it resolves more passes one by one and pays a matching error for each; raised, it leaves larger oscillations to be averaged. The residual in its three parts: the lag behind the glide; the matching error of [objective 4](#); and the averaging error of [objective 5](#). Each constant is to hold for every selection of the interval field, and its range of validity—bounded away from circular speed—is part of the statement.

Success. Formulas whose evaluation over the corridor in validated interval arithmetic bounds the distance from sampled trajectories of the full model to the reachable set of the limit system, with no fitted number.

3 The application: hypersonic entry

Entry is not an arbitrary illustration of the machinery above. It supplies the singular perturbation, the limit system and the concentrating events in one system, and the existing methods, for the reasons that follow, stop short of it.

3.1 Why existing methods do not reach it

The direct method solves a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in state dimension and capped in practice near four or five states [\[39, 40\]](#). A rigid-body entry vehicle carries thirteen states,

$$x = (r, \lambda, \phi, V, \gamma, \psi, q_0, q_1, q_2, q_3, p, q_b, r_b) \in \mathbb{R}^{13}, \quad (3.1)$$

and more once the heat load, the mass, the thermal response and recession of the heat shield and the aerothermoelastic modes are coupled [\[41\]](#). The two counts use “degree of freedom” in its two customary senses, and it is worth separating them: the vehicle is six-degree-of-freedom in the configuration sense (three translational, three rotational), while [Eq. \(3.1\)](#) has thirteen coordinates spanning a twelve-dimensional state manifold once the unit-quaternion constraint $\|\vec{q}\| = 1$ is imposed.

So the grid method stops before the real problem starts.

Set-propagation methods go further. Taylor-model and zonotope flowpipes return rigorous outer bounds at ten to twenty states, and they are the baseline this program has to be measured against [\[19, 42, 43\]](#). What they return is an enclosure of one instance: a numerical set for one vehicle, one initial box and one horizon, whose tightness tends to degrade with the width of the initial set and with the length of weakly damped motion it has to follow—and the phugoid is weakly damped for the whole of the glide. They are the right tool inside a pass, where the horizon is short, and that is how [objective 3](#) uses them. They do not produce a bound whose dependence on the corridor and on the interval widths is explicit, which is what a statement uniform over a family requires.

In practice the question is answered by sampling. Entry guidance has converged on numerical predictor–corrector algorithms whose performance is established by simulation over dispersed cases [\[38\]](#): a sample of the family, and not an enclosure of it.

What makes the measure-theoretic route viable is not that it evades the dimension but that the bound it produces does not scale with it: the limit system is low-dimensional by construction, each pass enters through one map, and the residual is a single ball.

3.2 The state space and the corridor

The translational state is carried in geocentric spherical position (r, λ, ϕ) and wind-frame velocity (V, γ, ψ) , so that the altitude and speed bounds of the entry corridor are box constraints on states and the heating, dynamic-pressure and load limits are monotone in the two coordinates they bind—the natural setting for interval contraction. Attitude is a unit quaternion, free of Euler-angle singularities. The chart has a price, which is stated rather than hidden: the longitude and heading rates are singular at the geographic poles, so the corridor is taken with $|\phi| \leq \phi_{\max} < \pi/2$, and a trajectory class that overflies a pole needs a second chart or a Cartesian formulation.

Because interval arithmetic evaluates transcendentals natively, the algebraic adjunctions that recast the system in polynomial form [44] are not needed on the main line; they survive only inside the moment-sum-of-squares cross-check. The state space therefore carries only the physical dimensions—spherical position, wind-frame velocity, attitude quaternion, body rates, thermal, ablation, and modal states—with no artificial embeddings, and the two auxiliary states of the polynomial formulation, the density root $y = \sqrt{\rho/\rho_0}$ and the wall-temperature surrogate, are gone. This is a design decision rather than an open problem, and its trade is stated once: transcendental dynamics in coordinates native to the corridor and the glide, against polynomial dynamics in which the corridor is a nonlinear function of the state.

Objective 1: Independent verification

Question. Is the certificate right, by evidence that does not share its derivation?

To carry out. Three checks, in the order of [Section 1.2](#). Symbolic verification of every derivation behind the constants. A moment-sum-of-squares outer approximation on a reduced polynomial model, as a cross-check that shares neither the decomposition nor the interval arithmetic. And a comparison on a common benchmark—a capsule skip entry reconstructed from published data—against Taylor-model flowpipes and against sampled trajectories of the full model, reported for each coordinate group rather than as a single number.

Success. Containment of every sample by the certified set, agreement with the cross-check where both apply, and a stated tightness against the baseline, coordinate by coordinate.

4 Plan of work

Seven concrete obligations carry the program; they are stated where each arises above and gathered here, in order, as the plan of work.

1. *Interval embedding, uniform through the singular limit:* what the occupation measures of the interval inclusion converge to, and the Liouville constraint, with its jump term, that the limit satisfies.
2. *Validated evaluation of the certificate:* interval evaluation, contraction and integration that discharge the certificate with every rounding error accounted for, and that enclose the reachable set of the limit system.
3. *Enclosures of the pass maps:* tight enclosures carried through a sequence of atmospheric passes without premature blow-up.
4. *Composition of passes:* the factorization of a trajectory through the pass maps and the flows between them, with an explicit matching remainder.

5. *Averaging over the oscillations, through resonance*: an averaging theorem with constants, uniform over the selections and valid where frequencies cross—the **critical path**, on which the reduction to a limit system and the closed-form bound rest.
6. *Closed-form constants*: the count K_δ and the residual $\rho(\varepsilon, \delta)$ from the physical corridor and the interval widths, free of any numerical solver.
7. *Independent verification*: symbolic checks of the derivations, a moment-sum-of-squares cross-check, and a benchmark against set-propagation tools and sampled trajectories.

Objective 1 states the limit; objectives 3 to 5 establish the decomposition; objective 6 reduces it to closed form; objectives 2 and 7 compute with it and check it. Objective 5 gates the rest.

4.1 Sequencing

The dependency structure suggests three phases rather than seven parallel tracks. The first establishes the embedding—objective 1, together with the certified envelope of the atmospheric density that it needs in order to be stated on a concrete inclusion. The second attacks the critical path: objective 5, which is where the program either works or does not, and which should therefore be reached early enough that a negative result is still recoverable. Its first case is small and self-contained, the planar oscillator of Eq. (2.4) with its damping, its drift and a bank history acting on it, and it can be started at once. Objectives 3 and 4 run alongside, since the matching argument and the enclosure geometry constrain each other. The third phase reduces the decomposition to closed form—objective 6—completes the validated evaluation of objective 2, and runs the symbolic and moment-sum-of-squares checks of objective 7 against the result. The validated evaluation is the one part of the program that a weaker theorem does not shorten.

Two retreat positions are worth identifying now. If objective 5 resists a proof where frequencies cross, the fallback is a corridor on which the bands are ordered—a floor on the dynamic pressure, a ceiling on the speed below circular—with the crossings at the edges of each pass left to the pass enclosures of objective 3, which integrate the coupled field and need no separation, and with the elastic modes treated as a widened interval. That weakens the residual without invalidating the decomposition. And if the amplitude of the oscillation about the glide cannot be bounded over every bank history, the statement is made for bank histories that vary slowly against the oscillation, for which the amplitude decays. That restricts the family and keeps the bound explicit.

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